PH7302: Condensed Matter Physics I

Instructor's Lecture Walkthrough & Comprehensive Notes

Module: Magnetism — Diamagnetism, Paramagnetism, Exchange Interaction, Ferromagnetism & Antiferromagnetism

Pacing: Two Lectures × 90 Minutes (Total: 3 Hours) | References: Prof. G. Rangarajan (NPTEL), Kittel, Ashcroft & Mermin

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LECTURE 1: Foundations, Diamagnetism, and Paramagnetism (90 Mins)

Pacing Schedule for Lecture 1

Time Block	Duration	Topic & Pedagogical Focus
00:00 – 00:15	15 min	Macroscopic Electrodynamics, Units (SI vs. CGS), Bohr–van Leeuwen Theorem
00:15 – 00:35	20 min	Atomic & Quantum Diamagnetism: Larmor Precession and Langevin Equation
00:35 – 01:05	30 min	Localized Paramagnetism: Hund's Rules, Brillouin Function, Curie Law, Van Vleck
01:05 – 01:25	20 min	Itinerant Paramagnetism: Conduction Electrons, Pauli Susceptibility, Landau Diamagnetism
01:25 – 01:30	05 min	Adiabatic Demagnetization & Lecture 1 Wrap-up / Practice Problems

1 Fundamental Definitions & The Quantum Origin of Magnetism (00:00 – 00:15)

1.1 Macroscopic Quantities and Unit Conventions

[Board Walkthrough Note]: Start by clarifying the notorious unit confusion in magnetism (Kittel uses CGS alongside SI). Write both side-by-side on the upper-right corner of the board and keep them visible.

- **Magnetization (M):** Magnetic dipole moment per unit volume:

$$\mathbf{M} = \frac{1}{V}\sum_i \boldsymbol{\mu}_i = -\frac{1}{V}\left(\frac{\partial F}{\partial \mathbf{B}}\right)_{T,V} \quad \text{or} \quad \mu_i = -\frac{\partial E_i}{\partial B} \tag{1}$$

  where $F = -k_B T \ln Z$ is the Helmholtz free energy.

- **Relation between B, H, and M:**

$$\text{SI:} \quad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}), \qquad \text{CGS:} \quad \mathbf{B} = \mathbf{H} + 4\pi\mathbf{M} \tag{2}$$

- **Magnetic Susceptibility (χ):** Dimensionless in both systems (per unit volume):

$$\text{SI:} \quad \chi = \frac{\partial M}{\partial H} \approx \frac{\mu_0 M}{B}, \qquad \text{CGS:} \quad \chi = \frac{\partial M}{\partial H} \approx \frac{M}{B} \tag{3}$$

  Note conversion: $\chi_{\text{SI}} = 4\pi\chi_{\text{CGS}}$. Molar susceptibility is $\chi_M = \chi \cdot (M_{\text{mol}}/\rho)$.

1.2 Why Magnetism is Strictly Quantum Mechanical: Bohr–van Leeuwen Theorem

[Instructor Prompt]: Ask students: *"If a magnetic field exerts a Lorentz force* $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$ *that curves electron trajectories into circles, why doesn't a classical gas of electrons show a net diamagnetic moment in thermal equilibrium?"*

Proof (Walkthrough on Board): In classical statistical mechanics, the canonical partition function for N particles of charge q_i in a vector potential $\mathbf{A}(\mathbf{r})$ (where $\mathbf{B} = \nabla \times \mathbf{A}$) is:

$$Z = \frac{1}{N!h^{3N}} \int d^{3N}r \int d^{3N}p \, \exp\left[-\frac{\beta}{2m}\sum_{i=1}^{N}\left(\mathbf{p}_i - q_i\mathbf{A}(\mathbf{r}_i)\right)^2 - \beta U(\mathbf{r}_1, \ldots, \mathbf{r}_N)\right] \tag{4}$$

For any fixed configuration $\{\mathbf{r}_i\}$, shift the momentum integration variable:

$$\mathbf{p}_i' = \mathbf{p}_i - q_i\mathbf{A}(\mathbf{r}_i) \implies d^3p_i' = d^3p_i \tag{5}$$

Because the limits of integration are $-\infty$ to $+\infty$, the shift eliminates $\mathbf{A}(\mathbf{r}_i)$ completely from the integral! Thus:

$$Z(\mathbf{B}) = Z(0) \implies F = -k_BT\ln Z \text{ is independent of } \mathbf{B} \implies \mathbf{M} = -\frac{1}{V}\frac{\partial F}{\partial \mathbf{B}} = \mathbf{0} \tag{6}$$

Key Conclusion: In classical physics, bulk orbital currents are exactly canceled by skipping orbits at the boundary. **Magnetism is 100% a quantum phenomenon** arising from quantized energy levels and intrinsic spin.

2 Atomic Diamagnetism: Classical & Quantum Formulations (00:15 – 00:35)

2.1 Classical Langevin Derivation via Larmor Theorem

Diamagnetism ($\chi < 0$) is the universal tendency of electrical charges to shield the interior of a body from an applied magnetic field (the microscopic analog of Lenz's law). Unlike classical free charges, an electron bound in a stationary atomic orbital maintains a persistent induced current as long as $\mathbf{B}$ is applied.

1. **Larmor Precession:** By Larmor's theorem, the motion of electrons of charge $-e$ and mass m around a central nucleus in a weak magnetic field $\mathbf{B} = B\hat{z}$ is identical to the unperturbed motion plus a superimposed precession at the **Larmor angular frequency**:

$$\omega_L = \frac{eB}{2m} \quad \text{(SI)}, \qquad \omega_L = \frac{eB}{2mc} \quad \text{(CGS)} \tag{7}$$

2. **Induced Current and Magnetic Moment:** For Z electrons per atom, the precession constitutes an induced current loop:

$$I = \text{charge} \times \text{revolutions per unit time} = (-Ze)\left(\frac{\omega_L}{2\pi}\right) = -\frac{Ze^2B}{4\pi m} \tag{8}$$

   The magnetic moment $\mu = I \cdot A$ of a loop of projected area $A = \pi\langle\rho^2\rangle = \pi\langle x^2 + y^2\rangle$ is:

$$\mu_{\text{ind}} = -\frac{Ze^2B}{4m}\langle\rho^2\rangle \tag{9}$$

3. **Spherical Symmetry:** For a spherically symmetric atom, the mean square radius is $\langle r^2\rangle = \langle x^2\rangle + \langle y^2\rangle + \langle z^2\rangle$ with $\langle x^2\rangle = \langle y^2\rangle = \langle z^2\rangle = \frac{1}{3}\langle r^2\rangle$. Thus $\langle\rho^2\rangle = \langle x^2 + y^2\rangle = \frac{2}{3}\langle r^2\rangle$.

4. **Langevin Diamagnetic Susceptibility:** For N atoms per unit volume:

$$\chi_{\text{dia}} = \frac{\mu_0 N\mu_{\text{ind}}}{B} = -\frac{\mu_0 NZe^2}{6m}\langle r^2\rangle \quad \text{(SI)}, \qquad \chi_{\text{dia}} = -\frac{NZe^2}{6mc^2}\langle r^2\rangle \quad \text{(CGS)} \tag{10}$$

2.2 Quantum Mechanical Derivation of the Magnetic Hamiltonian

In quantum mechanics, we replace canonical momentum $\mathbf{p} \to \mathbf{p} + e\mathbf{A}$ (for electron charge $q = -e$) and include the intrinsic spin Zeeman term. Choosing the symmetric gauge for a uniform field $\mathbf{B} = B\hat{z}$:

$$\mathbf{A} = \frac{1}{2}(\mathbf{B} \times \mathbf{r}) \implies A_x = -\frac{1}{2}yB, \quad A_y = \frac{1}{2}xB, \quad A_z = 0 \quad (\text{where } \nabla \cdot \mathbf{A} = 0) \tag{11}$$

The kinetic energy term in the Hamiltonian for a sum over atomic electrons i is:

$$\mathcal{H}_{\text{kin}} = \sum_i \frac{1}{2m}\left(\mathbf{p}_i + e\mathbf{A}(\mathbf{r}_i)\right)^2 = \sum_i \frac{p_i^2}{2m} + \frac{e}{2m}\sum_i (\mathbf{p}_i \cdot \mathbf{A}_i + \mathbf{A}_i \cdot \mathbf{p}_i) + \frac{e^2}{2m}\sum_i A_i^2 \tag{12}$$

Since $\nabla \cdot \mathbf{A} = 0$, $\mathbf{p}$ and $\mathbf{A}$ commute ($\mathbf{p} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{p}$). Using $\mathbf{r}_i \times \mathbf{p}_i = \hbar \mathbf{l}_i$:

$$\frac{e}{m}\sum_i \mathbf{A}_i \cdot \mathbf{p}_i = \frac{e}{2m}\sum_i (\mathbf{B} \times \mathbf{r}_i) \cdot \mathbf{p}_i = \frac{e}{2m}\mathbf{B} \cdot \sum_i (\mathbf{r}_i \times \mathbf{p}_i) = \frac{e\hbar}{2m}\mathbf{B} \cdot \mathbf{L} = \mu_B \mathbf{B} \cdot \mathbf{L} \tag{13}$$

where $\mu_B = \frac{e\hbar}{2m} = 9.274 \times 10^{-24}$ J/T (0.9274×10^{-20} erg/G) is the **Bohr magneton**, and $\mathbf{L}$ is the dimensionless orbital angular momentum operator ($\mathbf{L}_{\text{phys}} = \hbar \mathbf{L}$). Adding the spin magnetic energy $g_0 \mu_B \mathbf{B} \cdot \mathbf{S}$ (with $g_0 \approx 2.0023 \approx 2$) and expanding $A_i^2 = \frac{1}{4}B^2(x_i^2 + y_i^2)$:

$$\boxed{\mathcal{H}_{\text{mag}} = \mu_B(\mathbf{L} + g_0\mathbf{S}) \cdot \mathbf{B} + \frac{e^2B^2}{8m}\sum_{i=1}^{Z}(x_i^2 + y_i^2)} \tag{14}$$

- **First Term (Paramagnetic, $\propto B$):** Linear in B. In first-order perturbation theory, gives permanent dipole orientation. In second-order perturbation theory, gives Van Vleck paramagnetism.
- **Second Term (Diamagnetic, $\propto B^2$):** Quadratic in B. For a closed-shell atom ($\mathbf{L} = \mathbf{S} = \mathbf{J} = 0$), the first term vanishes identically. First-order perturbation shift of the ground state $|0\rangle$ is:

$$\Delta E_0 = \frac{e^2B^2}{8m}\sum_i \langle 0|x_i^2 + y_i^2|0\rangle = \frac{Ze^2B^2}{12m}\langle r^2\rangle \implies \mu = -\frac{\partial \Delta E_0}{\partial B} = -\frac{Ze^2\langle r^2\rangle}{6m}B \tag{15}$$

  recovering the exact Langevin diamagnetic formula!

[Board Example — Noble Gases & Hydrogen]: For ground-state atomic Hydrogen, $\psi_{1s}(r) = (\pi a_0^3)^{-1/2}e^{-r/a_0}$, one integrates $\langle r^2\rangle = 3a_0^2$. For noble gases, molar susceptibilities χ_M (in 10^{-6} cm^3/mol, CGS) grow monotonically with atomic radius and Z: He (-1.9), Ne (-7.2), Ar (-19.4), Kr (-28.0), Xe (-43.0).

3 Paramagnetism of Localized Moments (00:35 – 01:05)

3.1 Atomic Magnetic Moments, Landé g-Factor, and Hund's Rules

Paramagnetism ($\chi > 0$) occurs in systems with unpaired electrons: odd-electron atoms/defects (F-centers, Na vapor), free radicals, metals, and most importantly ions with partially filled inner shells ($3d$ transition metals, $4f$ lanthanides/rare earths, $5f$ actinides).

For a free atom/ion in the Russell–Saunders (L–S) coupling regime, the total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Because $\mathbf{L} + 2\mathbf{S}$ precesses rapidly around the conserved vector $\mathbf{J}$, we project the magnetic moment operator $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + 2\mathbf{S})$ onto $\mathbf{J}$:

$$\boldsymbol{\mu}_{\text{eff}} = -g_J\mu_B\mathbf{J}, \qquad g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{16}$$

where g_J is the **Landé g-factor** ($g_J = 1$ for pure orbital $S = 0$; $g_J = 2$ for pure spin $L = 0$).

Hund's Three Rules for Ground-State Multiplet $^{2S+1}L_J$:

1. **Maximize S:** Electrons occupy orbitals to maximize total spin $S = \sum m_s$ consistent with the Pauli exclusion principle (minimizes intra-atomic Coulomb repulsion via exchange / Fermi hole).
2. **Maximize L:** Subject to Rule 1, maximize total orbital angular momentum $L = \sum m_l$ (electrons rotate in the same direction, keeping them further apart).
3. **Spin-Orbit Coupling ($\lambda \mathbf{L} \cdot \mathbf{S}$):**
   - Less than half-filled shell ($\leq$ half): $J = |L - S|$
   - More than half-filled shell ($>$ half): $J = L + S$
   - Exactly half-filled shell (d^5 like Mn^{2+}, Fe^{3+} or f^7 like Gd^{3+}): $L = 0 \implies J = S$ (S-state ion).

3.2 Quantum Theory of Paramagnetism: The Brillouin Function

Consider N non-interacting magnetic ions per unit volume with total angular momentum quantum number J in a magnetic field $\mathbf{B} = B\hat{z}$. The $(2J+1)$-fold degenerate ground state splits into Zeeman levels indexed by $m_J \in \{-J, -J+1, \ldots, +J\}$:

$$E(m_J) = -\mu_z B = g_J \mu_B m_J B \tag{17}$$

Let us define the dimensionless ratio of magnetic to thermal energy:

$$x \equiv \frac{g_J \mu_B J B}{k_B T} \implies E(m_J) = \frac{m_J}{J} x k_B T \tag{18}$$

[Step-by-Step Derivation on Board]: The single-ion canonical partition function Z_1 is a finite geometric series with common ratio $r = e^{-x/J}$:

$$Z_1 = \sum_{m_J=-J}^{+J} \exp\left(-\frac{m_J x}{J}\right) = e^{x} + e^{x(J-1)/J} + \cdots + e^{-x} \tag{19}$$

Summing the geometric series of $2J+1$ terms and multiplying top and bottom by $e^{-x/(2J)}$:

$$Z_1 = \frac{e^{x} - e^{-x(J+1)/J}}{1 - e^{-x/J}} = \frac{e^{x(2J+1)/(2J)} - e^{-x(2J+1)/(2J)}}{e^{x/(2J)} - e^{-x/(2J)}} = \frac{\sinh\left(\frac{2J+1}{2J}x\right)}{\sinh\left(\frac{1}{2J}x\right)} \tag{20}$$

The thermal average magnetization $M = N\langle \mu_z \rangle$ is obtained from the free energy $F = -N k_B T \ln Z_1$:

$$M = -\frac{\partial F}{\partial B} = N k_B T \frac{\partial \ln Z_1}{\partial x}\frac{\partial x}{\partial B} = N g_J \mu_B J \frac{\partial}{\partial x}\left[\ln\sinh\left(\frac{2J+1}{2J}x\right) - \ln\sinh\left(\frac{1}{2J}x\right)\right] \tag{21}$$

Taking the derivative yields the celebrated **Curie–Brillouin Equation**:

$$\boxed{M = N g_J \mu_B J \, B_J(x), \qquad x = \frac{g_J \mu_B J B}{k_B T}} \tag{22}$$

where $B_J(x)$ is the **Brillouin Function**:

$$\boxed{B_J(x) = \frac{2J+1}{2J}\coth\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J}\coth\left(\frac{1}{2J}x\right)} \tag{23}$$

[FIGURE / SLIDE PLACEHOLDER 1 — Brillouin Function $B_J(x)$ vs. x]:

Board Sketch / Python Plot Space (See Appendix Script 1 — `fig1_brillouin.pdf`):
Draw axes: Vertical axis $M/M_{\text{sat}} = B_J(x)$ from 0 to 1.0; Horizontal axis $x = g_J\mu_B JB/k_BT$ from 0 to 6.
Sketch curves all starting at $(0, 0)$ and saturating asymptotically at 1.0:

- Top curve: $J = 1/2$ ($\tanh x$), steepest initial slope.
- Middle curves: $J = 3/2, 5/2, 7/2$ (Gd^{3+}).
- Bottom curve: Classical limit $J \to \infty$, Langevin function $L(x) = \coth x - 1/x$.

3.3 Limiting Cases of the Brillouin Function

1. **Two-Level Spin-1/2 System ($J = S = 1/2$, $L = 0$, $g_J = 2$):** Using $\coth(2x) - \frac{1}{2}\coth(x) = \tanh(x)$ (or directly from $Z_1 = e^x + e^{-x} = 2\cosh x$):

$$B_{1/2}(x) = \tanh(x) \implies M = N\mu_B \tanh\left(\frac{\mu_B B}{k_B T}\right) \tag{24}$$

2. **Classical Limit ($J \to \infty$):** When all orientations are allowed continuously, $\frac{2J+1}{2J} \to 1$ and $\frac{1}{2J}\coth\left(\frac{x}{2J}\right) \to \frac{1}{x}$:

$$\lim_{J\to\infty} B_J(x) = \coth(x) - \frac{1}{x} \equiv L(x) \quad \text{(Langevin Function)} \tag{25}$$

3. **Saturation Limit ($x \gg 1$, i.e., High Field / Very Low T):** Since $\coth(u) \to 1$ as $u \to \infty$, $B_J(x) \to \frac{2J+1}{2J} - \frac{1}{2J} = 1$, giving $M_{\text{sat}} = Ng_J\mu_B J$.

4. **Weak-Field / High-Temperature Limit ($x \ll 1$, Room Temperature & Lab Fields):** Note that at $T = 300$ K and $B = 1$ T, $\mu_B B/k_B T \approx 0.67$ K/300 K $\approx 2.2 \times 10^{-3} \ll 1$. Using the Taylor expansion $\coth(u) = \frac{1}{u} + \frac{u}{3} - \frac{u^3}{45} + \ldots$:

$$B_J(x) \approx \frac{2J+1}{2J}\left[\frac{2J}{(2J+1)x} + \frac{2J+1}{6J}x\right] - \frac{1}{2J}\left[\frac{2J}{x} + \frac{x}{6J}\right] = \frac{(2J+1)^2 - 1}{12J^2}x = \frac{J+1}{3J}x \tag{26}$$

Substituting $x = g_J\mu_B JB/k_BT$ into $M = Ng_J\mu_B JB_J(x)$ gives **Curie's Law**:

$$\boxed{\chi_{\text{Curie}} = \frac{\mu_0 M}{B} = \frac{\mu_0 N g_J^2 \mu_B^2 J(J+1)}{3k_B T} = \frac{\mu_0 N p_{\text{eff}}^2 \mu_B^2}{3k_B T} = \frac{C}{T}} \tag{27}$$

where C is the **Curie constant** and $p_{\text{eff}} = g_J\sqrt{J(J+1)}$ is the **effective Bohr magneton number**.

3.4 Rare Earth ($4f$) vs. Iron-Group ($3d$) Ions: Orbital Quenching

[Crucial Physical Comparison for Exam/Class]:

- **Rare Earth Ions ($4f^n$):** The $4f$ shell lies deep inside the ion, shielded by the filled $5s^2 5p^6$ outer shells. Crystal electric fields from neighboring ions are weak compared to spin-orbit coupling ($\mathcal{H}_{\text{CF}} \ll \lambda \mathbf{L}\cdot\mathbf{S}$). Thus J remains a good quantum number, and experimental p_{eff} agrees excellently with $g_J\sqrt{J(J+1)}$ (except Sm^{3+} and Eu^{3+}, where low-lying excited J-multiplets are thermally populated at 300 K).

- **Iron-Group Transition Metal Ions ($3d^n$):** The $3d$ electrons are on the outside of the ion and experience an intense, inhomogeneous electrostatic **crystal field** ($\mathcal{H}_{\rm CF} \sim 1$–$2$ eV $\gg \lambda \mathbf{L}\cdot\mathbf{S} \sim 0.01$–$0.05$ eV). The crystal field breaks the L–S coupling and splits the $(2L+1)$ orbital states into non-magnetic standing waves (e.g., real orbitals $d_{xy}, d_{x^2-y^2}$ in an octahedral or tetragonal field) for which $\langle L_z\rangle = 0$.

- **Result (Quenching of Orbital Angular Momentum):** Only the spin contributes to the magnetic moment! For $3d$ ions in solids:

$$\boxed{p_{\rm eff}(3d) \approx 2\sqrt{S(S+1)} = \sqrt{n(n+2)}} \quad \text{(where } n = \text{number of unpaired } d \text{ electrons)} \tag{28}$$

Table 1: Comparison of Theoretical and Experimental Effective Magneton Numbers ($p_{\rm eff}$)

Ion (Shell)	**Ground Term**	$g_J\sqrt{J(J+1)}$	$2\sqrt{S(S+1)}$	$p_{\rm exp}$	**Physical Regime**
Nd^{3+} $(4f^3)$	$^4I_{9/2}$	3.62	3.87	3.5	Unquenched ($4f$ shielded)
Gd^{3+} $(4f^7)$	$^8S_{7/2}$	7.94	7.94	8.0	Half-filled ($L=0$)
Dy^{3+} $(4f^9)$	$^6H_{15/2}$	10.63	5.92	10.6	Unquenched ($4f$ shielded)
Ti^{3+}, V^{4+} $(3d^1)$	$^2D_{3/2}$	1.55	1.73	1.8	Orbital Quenched ($3d$ exposed)
Cr^{3+} $(3d^3)$	$^4F_{3/2}$	0.77	3.87	3.8	Orbital Quenched ($3d$ exposed)
Mn^{2+}, Fe^{3+} $(3d^5)$	$^6S_{5/2}$	5.92	5.92	5.9	Half-filled ($L=0$)
Fe^{2+} $(3d^6)$	$^5D_4$	6.70	4.90	5.4	Orbital Quenched (partial SO)

3.5 Van Vleck Temperature-Independent Paramagnetism

Consider an atom/molecule whose ground state $|0\rangle$ is non-degenerate and has zero magnetic moment ($\langle 0|\mu_z|0\rangle = 0$, e.g., $J=0$ as in Eu^{3+} ground state $^7F_0$). In a magnetic field, $\mathcal{H}' = -\mu_z B$ mixes excited states $|s\rangle$ of energy $E_s - E_0 = \Delta_s \gg k_B T$ into the ground state via second-order perturbation theory:

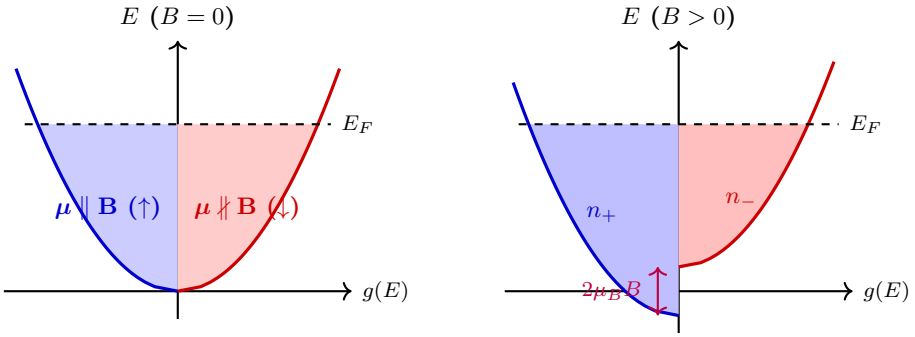
$$\Delta E_0^{(2)} = -\sum_{s\neq 0} \frac{|\langle s|\mu_z|0\rangle|^2 B^2}{E_s - E_0} \implies \chi_{\rm VV} = -\frac{\mu_0 N}{B}\frac{\partial \Delta E_0^{(2)}}{\partial B} = 2\mu_0 N \sum_{s\neq 0}\frac{|\langle s|\mu_z|0\rangle|^2}{E_s - E_0} > 0 \tag{29}$$

This gives a small, positive, **temperature-independent** susceptibility known as **Van Vleck paramagnetism**.

4 Paramagnetism of Conduction Electrons: Pauli Spin Susceptibility (01:05 – 01:25)

4.1 The Classical Puzzle vs. Fermi–Dirac Statistics

If conduction electrons in a metal behaved as free localized spins, they would obey Curie's law $\chi \sim N\mu_B^2/k_B T$, which diverges at low T. Experimentally, normal metals exhibit a weak, **temperature-independent** paramagnetism ($\sim 10^{-6}$). Wolfgang Pauli (1927) resolved this using Fermi–Dirac statistics: because of the Pauli exclusion principle, only electrons within a narrow thermal shell $\sim \mu_B B$ around the Fermi energy E_F can flip their spins!



(a) Unmagnetized ($n_\uparrow = n_\downarrow$) **(b) Zeeman Split** ($\Delta n \approx \mu_B B g(E_F)$)

Figure 1: Pauli spin paramagnetism of conduction electrons at $T = 0$. Applying a field $\mathbf{B}$ shifts the spin sub-bands by $\mp\mu_B B$, causing electrons within $\mu_B B$ of E_F to spill into the lower-energy sub-band.

4.2 Derivation of Pauli Paramagnetic Susceptibility

Let $g(E)$ be the total density of states (both spins) per unit volume, so each spin direction has $\frac{1}{2}g(E)$. At $T = 0$ (valid since $T \ll T_F \sim 5 \times 10^4$ K):

$$n_+ = \frac{1}{2}\int_{-\mu_B B}^{E_F} g(E + \mu_B B)\, dE \approx \frac{1}{2}\int_0^{E_F} g(E)\, dE + \frac{1}{2}\mu_B B\, g(E_F) \tag{30}$$

$$n_- = \frac{1}{2}\int_{+\mu_B B}^{E_F} g(E - \mu_B B)\, dE \approx \frac{1}{2}\int_0^{E_F} g(E)\, dE - \frac{1}{2}\mu_B B\, g(E_F) \tag{31}$$

The net concentration of excess aligned moments is $\Delta n = n_+ - n_- = \mu_B B g(E_F)$. Thus the magnetization is:

$$M = \mu_B(n_+ - n_-) = \mu_B^2 g(E_F) B \tag{32}$$

For a 3D free electron gas, recall $g(E_F) = \frac{3n}{2E_F} = \frac{3n}{2k_B T_F}$. Therefore, the **Pauli spin susceptibility** is:

$$\boxed{\chi_{\text{Pauli}} = \mu_0 \mu_B^2 g(E_F) = \frac{3\mu_0 n \mu_B^2}{2k_B T_F} = \chi_{\text{Curie}} \cdot \left(\frac{3T}{2T_F}\right)} \tag{33}$$

- **Landau Diamagnetism:** In a magnetic field, the orbital motion of conduction electrons is quantized into Landau levels, producing a diamagnetic response equal to exactly one-third of the Pauli spin susceptibility (for free electrons with effective mass m^*):

$$\chi_{\text{Landau}} = -\frac{1}{3}\left(\frac{m}{m^*}\right)^2 \chi_{\text{Pauli}} \implies \chi_{\text{total, free el.}} = \chi_{\text{Pauli}} + \chi_{\text{Landau}} = \frac{2}{3}\chi_{\text{Pauli}} \quad (\text{for } m^* = m) \tag{34}$$

4.3 Cooling by Adiabatic (Isentropic) Demagnetization (01:25 – 01:30)

For non-interacting spins, the magnetic entropy $S(T, B)$ depends only on the population distribution, which is a function solely of $x = \mu_B B / k_B T$, i.e., $S = S(B/T)$.

1. **Step 1 (Isothermal Magnetization at T_i):** Apply field B_i while in contact with a helium bath ($T_i \sim 1$ K). Spins align, entropy drops from $S(0)$ to $S(B_i/T_i)$, and heat $\Delta Q = T_i \Delta S$ is rejected to the bath.

2. **Step 2 (Adiabatic Demagnetization):** Thermally isolate the salt ($\Delta S = 0$) and slowly reduce $B_i \to B_{\text{local}}$ (where $B_{\text{local}} \sim 10^{-2}$ T is the internal dipolar field). Since $S(B/T) =$ const:

$$\frac{B_i}{T_i} = \frac{B_f}{T_f} \implies \boxed{T_f = T_i\left(\frac{B_{\text{local}}}{B_i}\right)} \tag{35}$$

allowing cooling down to millikelvin (10^{-3} K) with electronic spins, and microkelvin (μK) using nuclear spins (nuclear demagnetization of Cu).

LECTURE 2: Exchange Interaction, Ferromagnetism & Antiferromagnetism (90 Mins)

Pacing Schedule for Lecture 2

Time Block	Duration	Topic & Pedagogical Focus
00:00 – 00:20	20 min	Dipolar Failure & Quantum Exchange Interaction: Heitler–London H_2, Heisenberg Hamiltonian
00:20 – 00:45	25 min	Ferromagnetism: Weiss Molecular Field Theory, Curie–Weiss Law, Spontaneous Magnetization
00:45 – 01:05	20 min	Spin Waves (Magnons), Bloch $T^{3/2}$ Law, Domains, Bloch Walls & Hysteresis
01:05 – 01:30	25 min	Antiferromagnetism & Ferrimagnetism: 2-Sublattice Néel Model, $\chi_\parallel$ vs. $\chi_\perp$, Spin-Flop

5 The Exchange Interaction: Why Spins Order (00:00 – 00:20)

5.1 Why Magnetic Dipole–Dipole Interaction Cannot Explain Ferromagnetism

[Board Opener]: Iron has a Curie temperature $T_C = 1043$ K, meaning thermal energy $k_B T_C \approx 0.09$ eV is required to destroy spin alignment. Let us calculate the classical magnetic dipole–dipole interaction energy between two neighboring Bohr magnetons separated by lattice constant $a \approx 2$ Å:

$$U_{\text{dip}} \approx \frac{\mu_0 \mu_B^2}{4\pi a^3} \approx \frac{(10^{-7}\ \text{T}\cdot\text{m/A})(9.27\times10^{-24}\ \text{J/T})^2}{(2\times10^{-10}\ \text{m})^3} \approx 1.1{\times}10^{-24}\ \text{J} \approx 10^{-4}\ \text{eV} \implies T_{\text{dip}} = \frac{U_{\text{dip}}}{k_B} \approx 0.1\text{–}1\ \text{K} \tag{36}$$

Conclusion: Direct magnetic dipole coupling is **four orders of magnitude too weak** to explain room-temperature ferromagnetism! Instead, magnetic ordering arises from the **electrostatic Coulomb interaction combined with the Pauli Exclusion Principle** — the **Exchange Interaction**.

5.2 Two-Electron Toy Model (Heitler–London H_2) and the Heisenberg Hamiltonian

Consider two electrons at positions $\mathbf{r}_1, \mathbf{r}_2$ in orbitals ϕ_a, ϕ_b centered on sites a and b. The two-electron Hamiltonian $\mathcal{H} = \mathcal{H}_0 + \frac{e^2}{4\pi\epsilon_0 r_{12}}$ is completely spin-independent! However, the total wavefunction $\Psi(\mathbf{r}_1, s_1; \mathbf{r}_2, s_2) = \psi_{\text{space}}(\mathbf{r}_1, \mathbf{r}_2)\chi_{\text{spin}}(s_1, s_2)$ must be **antisymmetric** under simultaneous exchange $(1 \leftrightarrow 2)$:

1. **Spin Singlet ($S = 0$, Antiparallel Spins):** $\chi_{\text{singlet}} = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$ is antisymmetric, so the spatial part must be **symmetric** (bonding, charge piles up between nuclei):

$$\psi_S(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2(1+S_{ab}^2)}}\left[\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) + \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)\right], \qquad E_S = \frac{K + J_{\text{ex}}}{1 + S_{ab}^2} \approx K + J_{\text{ex}} \tag{37}$$

2. **Spin Triplet ($S = 1$, Parallel Spins):** $\chi_{\text{triplet}} = \{\uparrow\uparrow, \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow), \downarrow\downarrow\}$ is symmetric, so the spatial part must be **antisymmetric** (node at $\mathbf{r}_1 = \mathbf{r}_2$, creating a **Fermi hole** that keeps parallel-spin electrons apart):

$$\psi_T(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2(1-S_{ab}^2)}}\left[\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)\right], \qquad E_T = \frac{K - J_{\text{ex}}}{1 - S_{ab}^2} \approx K - J_{\text{ex}} \tag{38}$$

Here K is the direct Coulomb integral and J_{ex} is the **Exchange Integral**. Notice that the energy splitting $E_S - E_T = 2J$ depends purely on total spin $S \in \{0, 1\}$. Using the spin operator identity:

$$\mathbf{S}_{\text{tot}}^2 = (\mathbf{s}_1+\mathbf{s}_2)^2 = s_1^2+s_2^2+2\mathbf{s}_1\cdot\mathbf{s}_2 = \frac{3}{4}+\frac{3}{4}+2\mathbf{s}_1\cdot\mathbf{s}_2 \implies \mathbf{s}_1\cdot\mathbf{s}_2 = \frac{1}{2}S(S+1)-\frac{3}{4} = \begin{cases} -3/4 & (S=0 \text{ singlet}) \\ +1/4 & (S=1 \text{ triplet}) \end{cases} \tag{39}$$

We can write an effective spin Hamiltonian that reproduces both E_S and E_T identically:

$$\mathcal{H}_{\text{eff}} = \frac{1}{4}(E_S + 3E_T) - (E_S - E_T)\mathbf{s}_1 \cdot \mathbf{s}_2 = \text{const} - 2J\mathbf{s}_1 \cdot \mathbf{s}_2, \qquad J \equiv \frac{E_S - E_T}{2} \tag{40}$$

Generalizing to a crystal lattice of spins $\mathbf{S}_i$ yields the **Heisenberg Hamiltonian**:

$$\boxed{\mathcal{H}_{\text{Heis}} = -2\sum_{\langle i,j\rangle} J_{ij}\mathbf{S}_i \cdot \mathbf{S}_j - g\mu_B\mathbf{B}\cdot\sum_i \mathbf{S}_i} \quad (\text{or } -\sum_{i\neq j} J_{ij}\mathbf{S}_i \cdot \mathbf{S}_j) \tag{41}$$

- $J > 0 \implies E_T < E_S \implies$ **Ferromagnetic Coupling** (parallel spins $\uparrow\uparrow$ minimize energy).
- $J < 0 \implies E_S < E_T \implies$ **Antiferromagnetic Coupling** (antiparallel spins $\uparrow\downarrow$ minimize energy, as in covalent H_2 bond).

5.3 Types of Exchange Mechanisms in Solids

1. **Direct Exchange (Bethe–Slater Curve):** Direct overlap of neighboring d orbitals. Depends critically on the ratio r_a/r_{3d} (interatomic distance divided by radius of $3d$ shell). Small r_a/r_{3d} (Mn, Cr) gives strong overlap $\rightarrow$ covalent bonding $\rightarrow J < 0$ (antiferromagnetic). Intermediate r_a/r_{3d} (Fe, Co, Ni) gives weak overlap where intra-atomic Fermi-hole repulsion reduction wins $\rightarrow J > 0$ (ferromagnetic).

2. **Superexchange:** Indirect coupling of transition-metal cations across an intermediate nonmagnetic anion (e.g., Mn^{2+}–O^{2-}–Mn^{2+} in MnO). Usually gives strong antiferromagnetic coupling ($J < 0$).

3. **RKKY (Ruderman–Kittel–Kasuya–Yosida) Indirect Exchange:** In rare-earth metals (Gd, Tb, Dy), localized $4f$ spins polarize the surrounding sea of conduction electrons with an oscillatory spatial decay:

$$J_{\text{RKKY}}(r) \propto \frac{\cos(2k_F r)}{(2k_F r)^3} \tag{42}$$

leading to ferromagnetic, antiferromagnetic, or helical spin order depending on interatomic spacing.

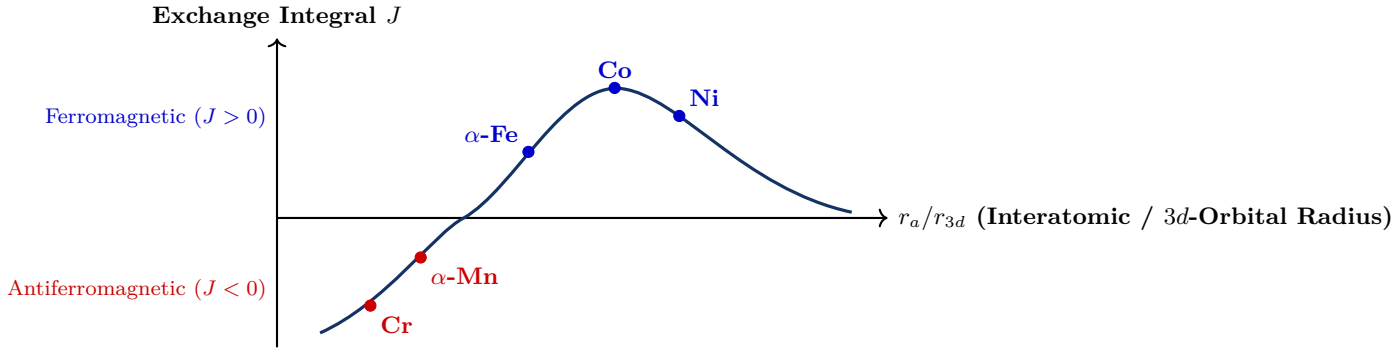


Figure 2: The Bethe–Slater curve illustrating the sign and magnitude of the direct exchange integral J as a function of the ratio r_a/r_{3d} for $3d$ transition metals.

6 Ferromagnetism & Weiss Molecular Field Theory (00:20 – 00:45)

6.1 From Heisenberg Hamiltonian to the Weiss Exchange Field

Consider site i in the Heisenberg Hamiltonian interacting with z nearest neighbors via exchange constant J:

$$\mathcal{H}_i = -2J\mathbf{S}_i \cdot \sum_{\delta=1}^{z} \mathbf{S}_{i+\delta} - g\mu_B \mathbf{S}_i \cdot \mathbf{B}_a \tag{43}$$

In the **Mean-Field (Weiss Molecular Field) Approximation**, we neglect correlated fluctuations $(\mathbf{S}_i - \langle\mathbf{S}\rangle)\cdot(\mathbf{S}_j - \langle\mathbf{S}\rangle)$ and replace the neighbor spins by their thermal average $\langle\mathbf{S}_{i+\delta}\rangle = \langle\mathbf{S}\rangle$. Since macroscopic magnetization is $\mathbf{M} = -Ng\mu_B\langle\mathbf{S}\rangle$:

$$\mathcal{H}_i^{\mathrm{MF}} = -g\mu_B \mathbf{S}_i \cdot (\mathbf{B}_a + \mathbf{B}_E) \implies \mathbf{B}_E = -\frac{2zJ}{g\mu_B}\langle\mathbf{S}\rangle = \left(\frac{2zJ}{Ng^2\mu_B^2}\right)\mathbf{M} \equiv \lambda\mathbf{M} \tag{44}$$

where λ is the **Weiss molecular field constant**:

$$\boxed{\mathbf{B}_{\mathrm{eff}} = \mathbf{B}_a + \lambda\mathbf{M}, \qquad \lambda = \frac{2zJ}{Ng^2\mu_B^2}} \tag{45}$$

[Board Estimate]: In Iron ($T_C \approx 1000$ K), the exchange field is $B_E = \lambda M_s \approx 10^3$ Tesla (10^7 Gauss), vastly exceeding any steady laboratory magnet!

6.2 Paramagnetic Regime ($T > T_C$): The Curie–Weiss Law

Above the critical temperature T_C, spontaneous magnetization vanishes, and a small applied field B_a induces a small magnetization M. Replacing $B \to B_a + \lambda M$ in the linear Curie law $M = \frac{C}{\mu_0 T}B_{\mathrm{eff}}$ (in CGS, $M = \frac{C}{T}(B_a + \lambda M)$):

$$MT = C(B_a + \lambda M) \implies M(T - C\lambda) = CB_a \implies \boxed{\chi = \frac{M}{B_a} = \frac{C}{T - T_C}} \tag{46}$$

This is the **Curie–Weiss Law**, where the ferromagnetic **Curie temperature** T_C is directly related to the microscopic exchange integral J:

$$\boxed{T_C = \lambda C = \left(\frac{2zJ}{Ng^2\mu_B^2}\right)\left(\frac{Ng^2\mu_B^2 S(S+1)}{3k_B}\right) = \frac{2zJS(S+1)}{3k_B}} \tag{47}$$

6.3 Ferromagnetic Regime ($T < T_C$): Spontaneous Magnetization $M_s(T)$

Set the external field $B_a = 0$, so $B_{\mathrm{eff}} = \lambda M$. Substituting B_{eff} into the exact Brillouin equation:

$$M(T) = M_0\, B_S(x), \qquad M_0 = Ng\mu_B S, \qquad x = \frac{g\mu_B S(\lambda M)}{k_B T} \tag{48}$$

Expressing $\lambda = \frac{3k_B T_C}{Ng^2\mu_B^2 S(S+1)}$, we eliminate M in favor of x:

$$\boxed{\frac{M(T)}{M_0} = B_S(x) = \left(\frac{S+1}{3S}\right)\left(\frac{T}{T_C}\right)x} \tag{49}$$

For the simplest case $S = 1/2$ ($g = 2$), $B_{1/2}(x) = \tanh(x)$ and $\frac{S+1}{3S} = 1$:

$$\boxed{m \equiv \frac{M(T)}{M_0} = \tanh\left(\frac{T_C}{T}m\right) \quad \Longleftrightarrow \quad \tanh(x) = \left(\frac{T}{T_C}\right)x} \tag{50}$$

[FIGURE / SLIDE PLACEHOLDER 4 — Graphical Solution of $M_s(T)$ & Universal Magnetization Curve]:

Board Sketch / Python Plot Space (See Appendix Script 2 — `fig2_weiss_ferro.pdf`):
Left Panel: Plot $y_1 = \tanh(x)$ and straight lines $y_2 = (T/T_C)x$ vs. x. Show that for $T > T_C$ (slope > 1), the only intersection is $x = 0$. For $T < T_C$ (slope < 1), a non-trivial intersection $x^* > 0$ exists!
Right Panel: Plot $M(T)/M(0)$ vs. T/T_C from 0 to 1.2. Curve starts flat at 1.0 at $T = 0$, drops smoothly, and plunges vertically into 0 at $T = T_C$ with critical exponent $\beta = 1/2$.

Asymptotic Behavior Near $T \to T_C^-$ (Critical Exponent $\beta = 1/2$): Near T_C, $m \ll 1$. Expanding $\tanh(u) \approx u - \frac{1}{3}u^3$:

$$m \approx \left(\frac{T_C}{T}\right) m - \frac{1}{3}\left(\frac{T_C}{T}\right)^3 m^3 \implies \frac{1}{3}\left(\frac{T_C}{T}\right)^3 m^2 = \frac{T_C - T}{T} \implies \boxed{\frac{M(T)}{M_0} \approx \sqrt{3}\left(1 - \frac{T}{T_C}\right)^{1/2}} \quad (51)$$

Asymptotic Behavior Near $T \to 0$ (Mean-Field Failure vs. Spin Waves): As $T \to 0$, $\tanh(x) \approx 1 - 2e^{-2x}$, so mean-field theory predicts an exponential deviation $M(T)/M_0 \approx 1 - 2\exp(-2T_C/T)$. **However, experiment shows a power law $\Delta M \propto T^{3/2}$!** Why? Because Mean-Field Theory assumes a single spin must flip completely against the entire exchange field ($\Delta E = 2zJS$), ignoring collective long-wavelength **spin waves**!

6.4 Band / Itinerant Ferromagnetism (Stoner Criterion for Fe, Co, Ni)

In $3d$ transition metals (Fe, Co, Ni), the $3d$ electrons are itinerant (delocalized in narrow energy bands) and have non-integer magnetic moments per atom (Ni: $0.60\,\mu_B$, Co: $1.72\,\mu_B$, Fe: $2.22\,\mu_B$). If we spontaneously transfer δn electrons from spin-down to spin-up bands at E_F:

- **Kinetic Energy Cost:** $\Delta E_{\text{kin}} = +\frac{1}{2}\frac{(\delta n)^2}{g(E_F)/2} = \frac{(\delta n)^2}{g(E_F)}$
- **Exchange Energy Gain:** $\Delta E_{\text{ex}} = -\frac{1}{2}I(\delta n)^2$ (where I is the Stoner exchange parameter).

Spontaneous band splitting ($\Delta E_{\text{tot}} < 0$) occurs when the **Stoner Criterion** is satisfied:

$$\boxed{I \cdot g(E_F) > 1} \quad (52)$$

This explains why Fe, Co, and Ni (where E_F sits inside a high-density $3d$ peak) are ferromagnetic, whereas Cu (where E_F sits in the low-density $4s$ band above the filled $3d$ band) is not, and Pd/Pt are "nearly ferromagnetic" Stoner-enhanced paramagnets.

7 Magnons (Spin Waves), Domains, and Hysteresis (00:45 – 01:05)

7.1 Ferromagnetic Magnons and Bloch's $T^{3/2}$ Law

In the ground state of a ferromagnet, all spins are parallel ($U_0 = -2NJS^2$ for a 1D chain). Reversing a single spin costs $\Delta E = 8JS^2$. However, a much lower energy excitation is obtained by tilting all spins slightly and letting them precess on cones with a constant phase advance ka between neighbors — a **spin wave** (quantized as a boson called a **magnon**).

1. **Equations of Motion:** Using $\frac{d\mathbf{S}_p}{dt} = \frac{1}{\hbar}(\mathbf{S}_p \times \text{Torque}) = -\frac{2J}{\hbar}\mathbf{S}_p \times (\mathbf{S}_{p-1} + \mathbf{S}_{p+1})$ and linearizing ($S_p^z \approx S$, $S_p^x, S_p^y \ll S$) for a plane wave $\propto e^{i(pka-\omega t)}$ yields the **Ferromagnetic Magnon Dispersion Relation**:

$$\boxed{\hbar\omega_k = 4JS(1-\cos ka) \xrightarrow{ka\ll 1} 2JSa^2k^2 \equiv Dk^2} \quad (\text{3D cubic: } \hbar\omega_k = 2JSa^2k^2) \tag{53}$$

Notice that ferromagnetic magnons have a **quadratic dispersion** ($\omega \propto k^2$) at long wavelengths!

2. **Bloch's $T^{3/2}$ Law:** Each thermally excited magnon reduces the total spin projection by 1 ($\Delta M = g\mu_B \sum_k n_k$). Since magnons obey Bose–Einstein statistics with zero chemical potential:

$$\sum_k n_k = \int \frac{d^3k}{(2\pi)^3} \frac{1}{e^{\hbar\omega_k/k_BT}-1} = \frac{1}{2\pi^2}\int_0^\infty \frac{k^2\,dk}{e^{Dk^2/k_BT}-1} \propto \left(\frac{k_BT}{D}\right)^{3/2} \tag{54}$$

Evaluating the Riemann Zeta integral gives **Bloch's $T^{3/2}$ Law** for both fractional magnetization drop and magnon heat capacity:

$$\boxed{\frac{\Delta M(T)}{M(0)} = \frac{M(0)-M(T)}{M(0)} = \frac{0.0587}{SQ}\left(\frac{k_BT}{2JS}\right)^{3/2} \propto T^{3/2}, \qquad C_{\text{mag}} \propto T^{3/2}} \tag{55}$$

where $Q = 1, 2, 4$ for simple cubic, bcc, and fcc lattices respectively.

7.2 Ferromagnetic Domains, Bloch Walls, and Hysteresis

Why isn't every piece of iron at room temperature ($T \ll T_C$) a strong permanent magnet? Because a uniformly magnetized block creates a huge external stray magnetic field $\mathbf{B}_{\text{stray}}$ with magnetostatic energy $\int \frac{B^2}{2\mu_0} dV$. To minimize magnetostatic energy, the crystal subdivides into macroscopic **Weiss Domains** with closure flux patterns.

Thickness and Energy of a Bloch Wall: The boundary between two antiparallel domains (180° rotation) is a **Bloch Wall**. Why doesn't the spin reverse abruptly in a single atomic step?

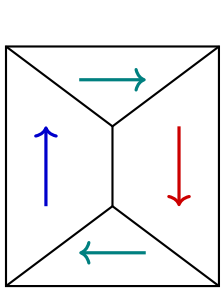
- Between two spins tilted by small angle ϕ, the exchange cost is $w_{\text{ex}} = -2JS^2\cos\phi - (-2JS^2) \approx JS^2\phi^2$.
- If a total rotation of π radians is spread over N atomic planes, $\phi = \pi/N$, so the total exchange energy of a line of N spins is:

$$E_{\text{ex, line}} = N \cdot JS^2\left(\frac{\pi}{N}\right)^2 = \frac{\pi^2JS^2}{N} \tag{56}$$

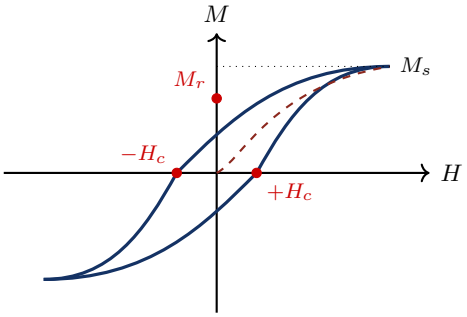
Thus exchange energy favors $N \to \infty$ (infinitely thick wall)!

- What stops the wall from growing infinitely wide? **Magnetocrystalline Anisotropy Energy (K)** (arising from spin-orbit coupling, which locks spins along easy crystallographic axes like [100] in Fe or c-axis in Co). Spins inside the wall point along hard directions, costing anisotropy energy per unit area $\sigma_{\text{anis}} \approx KNa$.
- Minimizing total wall energy per unit area $\sigma_{\text{wall}} = \sigma_{\text{ex}} + \sigma_{\text{anis}}$ with respect to N:

$$\sigma_{\text{wall}}(N) = \frac{\pi^2JS^2}{Na^2} + KNa, \qquad \frac{\partial\sigma_{\text{wall}}}{\partial N} = 0 \implies \boxed{N = \sqrt{\frac{\pi^2JS^2}{Ka^3}} \sim 300 \text{ planes } (\delta \sim 100 \text{ nm})} \tag{57}$$



(a) Flux-Closure Weiss Domains



(b) Ferromagnetic M–H Hysteresis Loop

Figure 3: **(a)** Flux-closure domain configuration minimizing external stray magnetostatic energy. **(b)** Ferromagnetic hysteresis loop showing virgin magnetization curve, saturation M_s, remanence M_r, and coercive field H_c.

8 Antiferromagnetism & Ferrimagnetism (01:05 – 01:30)

8.1 Two-Sublattice Néel Mean-Field Model for Antiferromagnets

In an **antiferromagnet** ($J < 0$, e.g., $\mathrm{MnO}, \mathrm{FeO}, \mathrm{CoO}, \mathrm{NiO}, \mathrm{MnF_2}, \mathrm{Cr}$), a bipartite crystal lattice splits into two interpenetrating sublattices A (spins ↑, magnetization $\mathbf{M}_A$) and B (spins ↓, magnetization $\mathbf{M}_B$) with $|\mathbf{M}_A| = |\mathbf{M}_B|$ below the **Néel temperature** T_N, giving **zero net spontaneous magnetization** ($\mathbf{M}_{\text{tot}} = \mathbf{M}_A + \mathbf{M}_B = 0$).

- **Experimental Detection via Neutron Diffraction (Shull & Smart, 1949):** In MnO ($T_N = 116$ K), X-ray diffraction shows a chemical FCC unit cell of $a = 4.43$ Å at both 293 K and 80 K. However, neutron diffraction (which senses magnetic moments) at 80 K reveals extra half-integer superlattice Bragg peaks $(1/2, 1/2, 1/2)$ corresponding to a magnetic unit cell of $2a = 8.85$ Å!

Derivation of Susceptibility Above T_N ($T > T_N$): Assuming dominant antiferromagnetic coupling between A and B sublattices (and neglecting intra-sublattice coupling for simplicity), the molecular fields acting on A and B are:

$$\mathbf{B}_A = \mathbf{B}_a - \mu\mathbf{M}_B, \qquad \mathbf{B}_B = \mathbf{B}_a - \mu\mathbf{M}_A \quad (\text{with } \mu > 0) \tag{58}$$

In the paramagnetic phase ($T > T_N$), each sublattice has Curie constant C (where $C_{\text{total}} = 2C$):

$$M_A T = C(B_a - \mu M_B), \qquad M_B T = C(B_a - \mu M_A) \tag{59}$$

1. **Néel Temperature T_N:** At $B_a = 0$, a non-trivial staggered solution $M_A = -M_B \neq 0$ requires:
$$\det \begin{pmatrix} T & \mu C \\ \mu C & T \end{pmatrix} = 0 \implies T^2 - (\mu C)^2 = 0 \implies \boxed{T_N = \mu C} \tag{60}$$

2. **Paramagnetic Susceptibility ($T > T_N$):** Adding the two equations gives $(M_A + M_B)T = 2CB_a - \mu C(M_A + M_B)$:
$$\boxed{\chi = \frac{M_A + M_B}{B_a} = \frac{2C}{T + \mu C} = \frac{2C}{T + T_N}} \quad \left(\text{Generally with } AA/BB \text{ coupling: } \chi = \frac{2C}{T + \Theta}\right) \tag{61}$$
Notice the **plus sign** in the denominator! Extrapolating $1/\chi$ vs. T from high temperatures intercepts the temperature axis at a **negative temperature** $T = -\Theta$ (or $-T_N$).

8.2 Anisotropic Susceptibility Below T_N: $\chi_\perp$ vs. $\chi_\parallel$ and Spin-Flop

Below T_N, the magnetic response of a single-crystal antiferromagnet depends strongly on the orientation of the applied field $\mathbf{B}_a$ relative to the easy axis of sublattice magnetization:

1. **Perpendicular Field ($\mathbf{B}_a \perp$ easy axis at $T = 0$):** The field tilts both $\mathbf{M}_A$ and $\mathbf{M}_B$ toward $\mathbf{B}_a$ by a small angle ϕ (so the angle between $\mathbf{M}_A$ and $\mathbf{M}_B$ is $\pi - 2\phi$, where $\cos(\pi - 2\phi) = -\cos(2\phi) \approx -(1 - 2\phi^2)$). The energy density is:

$$U = \mu \mathbf{M}_A \cdot \mathbf{M}_B - \mathbf{B}_a \cdot (\mathbf{M}_A + \mathbf{M}_B) \approx -\mu M^2(1 - 2\phi^2) - 2B_a M\phi \tag{62}$$

   Minimizing with respect to ϕ:

$$\frac{\partial U}{\partial \phi} = 4\mu M^2\phi - 2B_a M = 0 \implies \phi = \frac{B_a}{2\mu M} \implies \boxed{\chi_\perp(0) = \frac{2M\phi}{B_a} = \frac{1}{\mu} = \frac{2C}{2T_N} = \chi(T_N) = \text{const}} \tag{63}$$

   Remarkably, $\chi_\perp(T)$ is **independent of temperature** from $T = 0$ up to T_N!

2. **Parallel Field ($\mathbf{B}_a \parallel$ easy axis at $T = 0$):** At $T = 0$ K, sublattices A and B are completely saturated along $\pm\hat{z}$. A small parallel field lowers the energy of A by $-MB_a$ and raises the energy of B by $+MB_a$, resulting in zero net energy change and zero magnetization change:

$$\boxed{\chi_\parallel(0) = 0} \tag{64}$$

   As T increases from 0 to T_N, thermal agitation allows spins to flip, so $\chi_\parallel(T)$ rises smoothly from 0 to meet $\chi_\perp$ at T_N (creating the characteristic **cusp** at T_N).

3. **Polycrystalline (Powder) Average at $T = 0$ K:** Averaging over all grain orientations ($\frac{1}{3}$ parallel, $\frac{2}{3}$ perpendicular):

$$\boxed{\chi_{\text{poly}}(0) = \frac{1}{3}\chi_\parallel(0) + \frac{2}{3}\chi_\perp(0) = \frac{2}{3}\chi(T_N)} \tag{65}$$

4. **Spin-Flop Transition:** If $B_a \parallel$ easy axis is increased to a critical field $B_{\text{SF}} = \sqrt{2B_E B_A}$ (where B_A is the anisotropy field), the system abruptly flops its spins perpendicular to $\mathbf{B}_a$ to take advantage of the lower magnetic energy $-\frac{1}{2}\chi_\perp B_a^2 < -\frac{1}{2}\chi_\parallel B_a^2$!

5. **Antiferromagnetic Magnons:** Solving the coupled equations of motion for sublattices A and B gives a **linear dispersion relation** (just like acoustic phonons!):

$$\boxed{\hbar\omega_k = 4|J|S|\sin(ka)| \xrightarrow{ka \ll 1} 4|J|Sak \propto k} \implies C_{\text{mag, AFM}} \propto T^3 \tag{66}$$

[FIGURE / SLIDE PLACEHOLDER 6 — Master Comparison of $\chi(T)$ and $1/\chi(T)$]:
Board Sketch / Python Plot Space (See Appendix Script 3 — `fig3_susceptibility_comparison.pdf`):
Left Panel (χ vs. T): Show Ferromagnet diverging at T_C; Paramagnet decaying as $1/T$; Antiferromagnet showing a sharp cusp at T_N with three branches below T_N: $\chi_\perp$ (flat), χ_{poly} (drops to 2/3), and $\chi_\parallel$ (drops to 0).
Right Panel ($1/\chi$ vs. T): Show three straight lines: Paramagnet through $(0, 0)$, Ferromagnet intercepting at $+T_C$, Antiferromagnet intercepting at $-\Theta$, plus the curved hyperbolic branch of a Ferrimagnet plunging to T_C.

8.3 Ferrimagnetism (Magnetite Fe_3O_4 and Ferrites)

In a **ferrimagnet** (Néel, 1948), two sublattices A and B are coupled antiferromagnetically ($B_A = B_a - \mu M_B$, $B_B = B_a - \mu M_A$), but have **unequal magnetic moments** ($C_A \neq C_B$, $M_A \neq M_B$), resulting in a net spontaneous magnetization $M_{\text{net}} = |M_B - M_A| \neq 0$!

- **Classic Example — Magnetite ($\mathbf{Fe_3O_4 = Fe^{2+}O \cdot Fe_2^{3+}O_3}$, Inverse Spinel):** In one formula unit, there are two Fe^{3+} ions ($3d^5, S = 5/2, L = 0 \implies 5\,\mu_B$ each) and one Fe^{2+} ion ($3d^6, S = 2, L$ quenched $\implies 4\,\mu_B$). If all spins were parallel, M_{sat} would be $5+5+4 = 14\,\mu_B$ per formula unit. Experimentally, $M_{\text{sat}} = 4.1\,\mu_B$! **Why?** One Fe^{3+} ($5\,\mu_B$, $\downarrow$) occupies tetrahedral A-sites, while the other Fe^{3+} ($5\,\mu_B$, $\uparrow$) and the Fe^{2+} ($4\,\mu_B$, $\uparrow$) occupy octahedral B-sites. The two Fe^{3+} moments cancel completely ($+5-5 = 0$), leaving only the Fe^{2+} moment of $4\,\mu_B$!

- **Susceptibility of a Ferrimagnet ($T > T_C$):** Solving $M_A T = C_A(B_a - \mu M_B)$ and $M_B T = C_B(B_a - \mu M_A)$ gives:

$$\boxed{\chi = \frac{M_A + M_B}{B_a} = \frac{(C_A + C_B)T - 2\mu C_A C_B}{T^2 - T_C^2}, \qquad T_C = \mu\sqrt{C_A C_B}} \tag{67}$$

  which produces a characteristic downward-curving hyperbola in $1/\chi$ vs. T near T_C. Because ferrites ($MO \cdot Fe_2O_3$) are electrical insulators, they suppress eddy-current losses and are indispensable in high-frequency RF/microwave applications.

Master Summary Table for Classroom Review

Class of Material	Sign/Mag. of χ	Temp. Dependence	Microscopic Origin & Examples
Diamagnet	Small < 0 (-10^{-6})	Independent of T	Orbital Larmor shielding of closed shells (He, Ne, H_2O, Cu core, Bi).
Curie Paramagnet	Positive (10^{-4}–10^{-2})	$\chi = \dfrac{C}{T}$	Permanent localized moments ($3d, 4f$ salts, Gd^{3+}, O_2).
Pauli Paramagnet	Small > 0 (10^{-6}–10^{-5})	Independent of T	Spin-split conduction electrons near E_F in metals (Na, Al, Mg).
Ferromagnet	Huge > 0 (10^{2}–10^{5})	$\chi = \dfrac{C}{T - T_C}$ $(T > T_C)$	Parallel alignment ($J > 0$); spontaneous M_s below T_C (Fe, Co, Ni, Gd, EuO).
Antiferromagnet	Small > 0 (10^{-4}–10^{-3})	$\chi = \dfrac{2C}{T + \Theta}$ $(T > T_N)$	Equal antiparallel sublattices ($J < 0$); cusp at T_N (MnO, NiO, MnF_2, Cr).
Ferrimagnet	Large > 0	Hyperbolic $1/\chi$ $(T > T_C)$	Unequal antiparallel sublattices ($M_A \neq M_B$); insulating ferrites (Fe_3O_4, YIG).

In-Class / Homework Tutorial Problems (With Brief Solutions)

1. **Atomic Hydrogen Diamagnetism:** Using $\psi_{1s}(r) = (\pi a_0^3)^{-1/2} e^{-r/a_0}$, show that $\langle r^2 \rangle = 3a_0^2$ and calculate the molar diamagnetic susceptibility of atomic hydrogen.
   Solution: $\langle r^2 \rangle = \frac{4\pi}{\pi a_0^3} \int_0^\infty r^4 e^{-2r/a_0} dr = \frac{4}{a_0^3} \frac{4!}{(2/a_0)^5} = 3a_0^2$. Substituting into Langevin's formula gives $\chi_M = -\frac{\mu_0 N_A e^2 a_0^2}{2m} \approx -2.98 \times 10^{-11}$ m^3/mol (-2.37×10^{-6} cm^3/mol in CGS).

2. **Hund's Rules & Quenching in Fe^{2+} vs. Fe^{3+}:** Find the ground-state spectroscopic term $^{2S+1}L_J$, g_J, p_{eff} (unquenched), and p_{eff} (orbital quenched) for Fe^{2+} ($3d^6$) and Fe^{3+} ($3d^5$).
   Solution: For Fe^{3+} ($3d^5$): $S = 5/2, L = 0, J = 5/2 \implies {}^6S_{5/2}$, $g_J = 2$. Both formulas give $p_{\text{eff}} = 2\sqrt{(5/2)(7/2)} = \sqrt{35} = 5.92$ ($p_{\text{exp}} = 5.9$). For Fe^{2+} ($3d^6$): $S = 2, L = 2$, shell $>$ half-full $\implies J = L + S = 4 \implies {}^5D_4$, $g_J = 3/2$. Unquenched $p_{\text{eff}} = 1.5\sqrt{20} = 6.71$; Quenched $p_{\text{eff}} = 2\sqrt{2(3)} = \sqrt{24} = 4.90$ ($p_{\text{exp}} = 5.4$, confirming orbital quenching).

3. **Two-Level Paramagnetic Heat Capacity (Schottky Anomaly):** For a spin-1/2 paramagnet with $E_\pm = \mp \mu_B B$, show that the internal energy is $U = -N\mu_B B \tanh(\mu_B B / k_B T)$ and derive the magnetic specific heat $C_B = (\partial U / \partial T)_B$.
   Solution: Differentiating with respect to T gives $C_B = N k_B \left(\frac{\mu_B B}{k_B T}\right)^2 \operatorname{sech}^2\left(\frac{\mu_B B}{k_B T}\right)$, which peaks at $\mu_B B / k_B T \approx 1.2$ and decays as $1/T^2$ at high T.

4. **Exchange Integral in Iron:** Given $T_C = 1043$ K for bcc Iron ($z = 8$ nearest neighbors) and assuming $S = 1$, calculate the exchange integral J in meV and the Weiss field B_E.
   Solution: From $T_C = \frac{2zJS(S+1)}{3k_B}$, we get $J = \frac{3k_B T_C}{2(8)(1)(2)} = \frac{3}{32}(89.9 \text{ meV}) \approx 8.43$ meV. The molecular field at $T = 0$ is $B_E = \frac{2zJS}{g\mu_B} = \frac{3k_B T_C}{(S+1) g \mu_B} \approx 1160$ Tesla.